

RESPIRATORY DISEASES

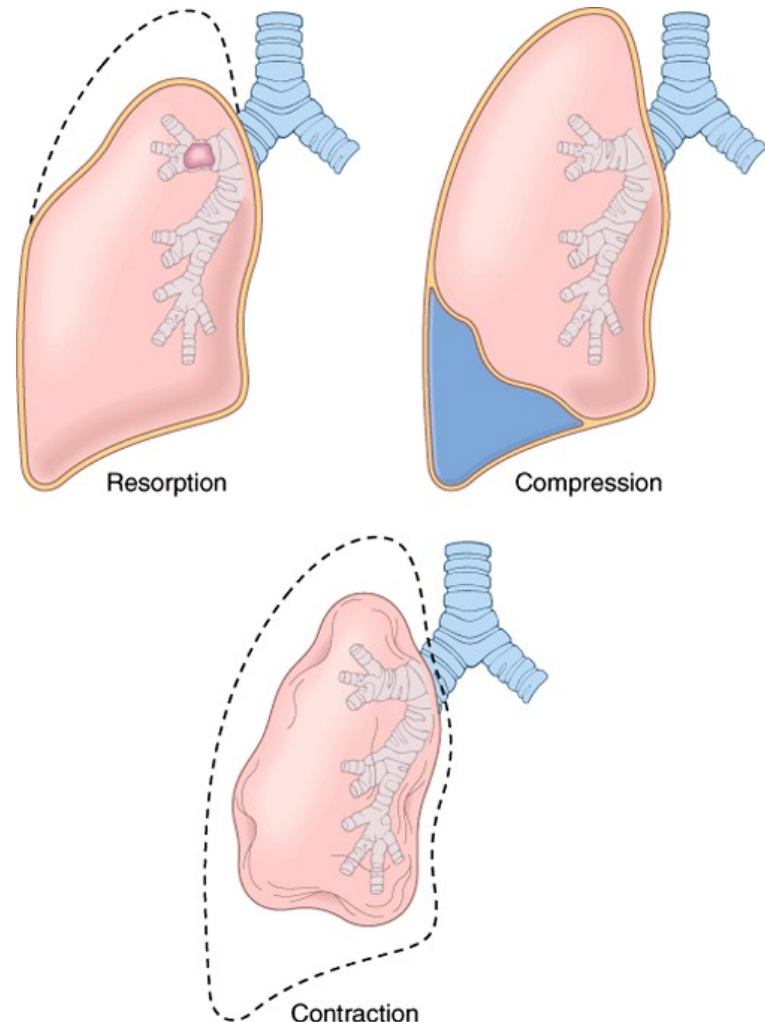
BY MELESE A. (MD)

OUTLINES

- Atelectasis
- Obstructive lung diseases
- Restrictive lung diseases
- Pneumonia

ATELECTASIS (COLLAPSE)

- Atelectasis refers either to
 - ✓ **incomplete expansion** of the lungs (**neonatal** atelectasis) or
 - ✓ the collapse of previously inflated lung, producing areas of relatively airless pulmonary parenchyma
- It results in shunting of inadequately oxygenated blood from pulmonary arteries into veins, thus giving rise to **hypoxia**
- Three types;
 - ✓ Resorption atelectasis
 - ✓ Compression atelectasis
 - ✓ Contraction atelectasis (cicatrization atelectasis)



1. RESORPTION ATELECTASIS

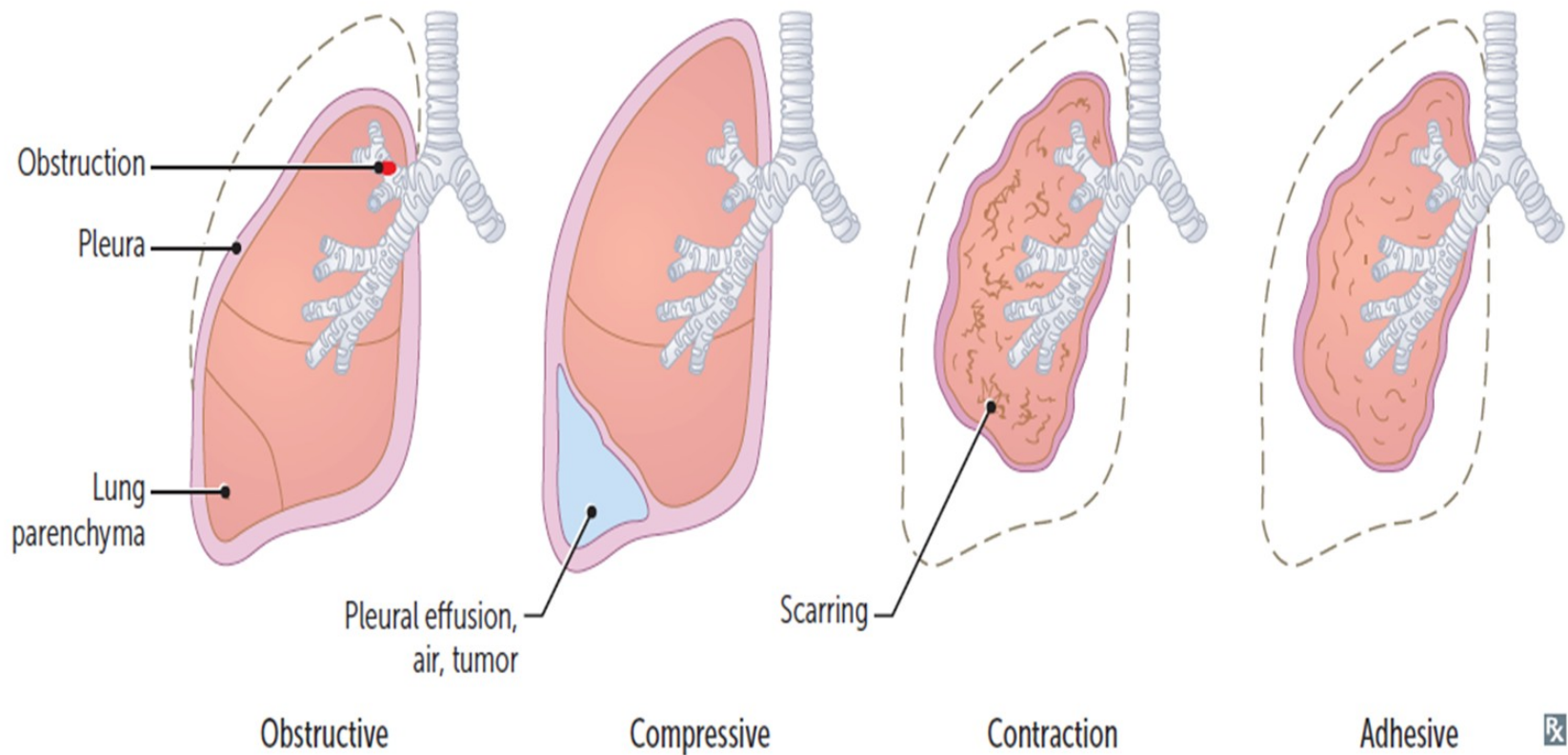
- Due to total obstruction of a bronchus preventing air from reaching distal airways
- The air already present gradually becomes absorbed, and alveolar collapse follows.
- mediastinum shifts toward the atelectatic lung.
- Caused by Obstruction of a bronchus by:
 - ✓ Intrabronchial mucous or mucopurulent plugs in post operative patients
 - ✓ Foreign body aspiration, especially in children
 - ✓ Obstructive lung disease: bronchial asthma, bronchiectasis, chronic bronchitis
 - ✓ Tumors.

2. COMPRESSION ATELECTASIS

- Caused by accumulation of fluid, blood, or air within pleural cavity, which mechanically collapse adjacent lung.
 - a. Pleural effusion
 - b. Pneumothorax: air in the pleural cavity
 - c. failure to breath deeply: bedridden patients, ascites, and during and after surgery
- Mediastinum shifts **away from** the affected lung.

3. CONTRACTION ATELECTASIS (CICATRIZATION ATELECTASIS)

- Occurs due to local or generalized **fibrosis** of the lung or pleura that prevents full expansion of the lung.
- Atelectasis (except when caused by **contraction**) is potentially reversible and should be treated promptly to prevent hypoxemia and superimposed infection of the collapsed lung.



OBSTRUCTIVE VS. RESTRICTIVE:

- Diffuse pulmonary diseases can be classified into **two** Categories:

1 - **Obstructive airway diseases:**

- characterized by an **increase in resistance to airflow** caused by partial or complete obstruction at any level
- Total lung capacity and forced vital capacity (FVC) are **NORMAL**, but Forced Expiratory Volume (FEV₁) is decreased (i.e. FEV:FVC ratio **REDUCED**)
- The major diffuse obstructive disorders are
 - ✓ **emphysema,**
 - ✓ **chronic bronchitis,**
 - ✓ **bronchiectasis, and**
 - ✓ **asthma**

2. Restrictive diseases:

- Characterized by reduced expansion of lung parenchyma and decreased total lung capacity.
- FVC and FEV₁ reduced proportionately
- **Restrictive** defects occur in **two** general conditions:
 - A. Chest wall disorders** in the presence of normal lungs:
 - ✓ severe obesity
 - ✓ diseases of the pleura
 - ✓ neuromuscular disorders that affect the respiratory muscles
 - B. Acute or chronic interstitial lung diseases:**
 - ✓ The classic **acute** restrictive disease is **ARDS**
 - ✓ Chronic restrictive diseases include the pneumoconioses, interstitial fibrosis of unknown etiology, and sarcoidosis

OBSTRUCTIVE LUNG DISEASES

- Characterized by **limitation of airflow** usually resulting from an **increase in resistance** caused by partial or complete **obstruction** at any level.
- Four disorders in this group:
 - ✓ **emphysema**
 - ✓ **chronic bronchitis**
 - ✓ **asthma**
 - ✓ **bronchiectasis**

1. EMPHYSEMA

- Abnormal **permanent** enlargement of the airspaces distal to the **terminal bronchioles** with **destruction of their walls** and **no fibrosis**
- Imbalance of **protease/antiprotease activity** and/or excess of reactive oxygen species activity
 - Excessive activation of neutrophil and macrophage **elastase** associated with **chronic smoking**
 - Hereditary **α_1 anti-trypsin** deficiency
- Dyspnea and hyperventilation but with adequate gas exchange (until very late in disease), aka **“pink puffers”**

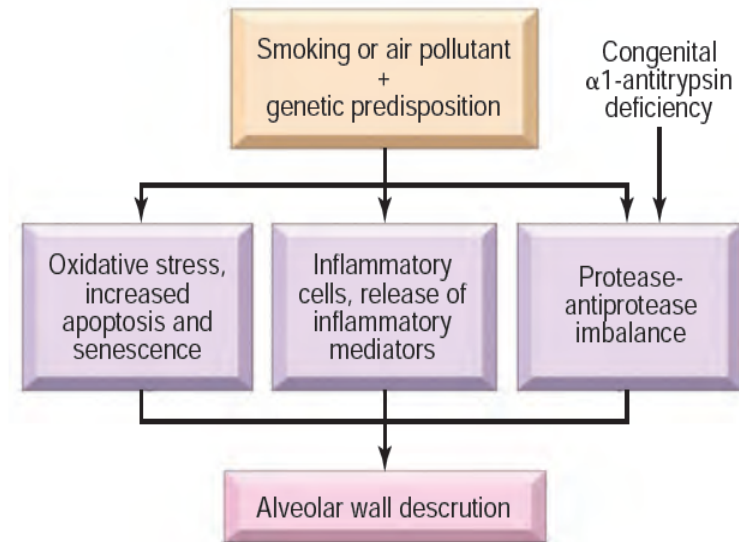


Figure 15-8 Pathogenesis of emphysema. See text for details.

- Classified according to it's **anatomic** distribution

a. Centriacinar (Centrilobular) Emphysema

- ✓ Most common
- ✓ Caused by **cigarette smoking**
- ✓ **upper lobes**

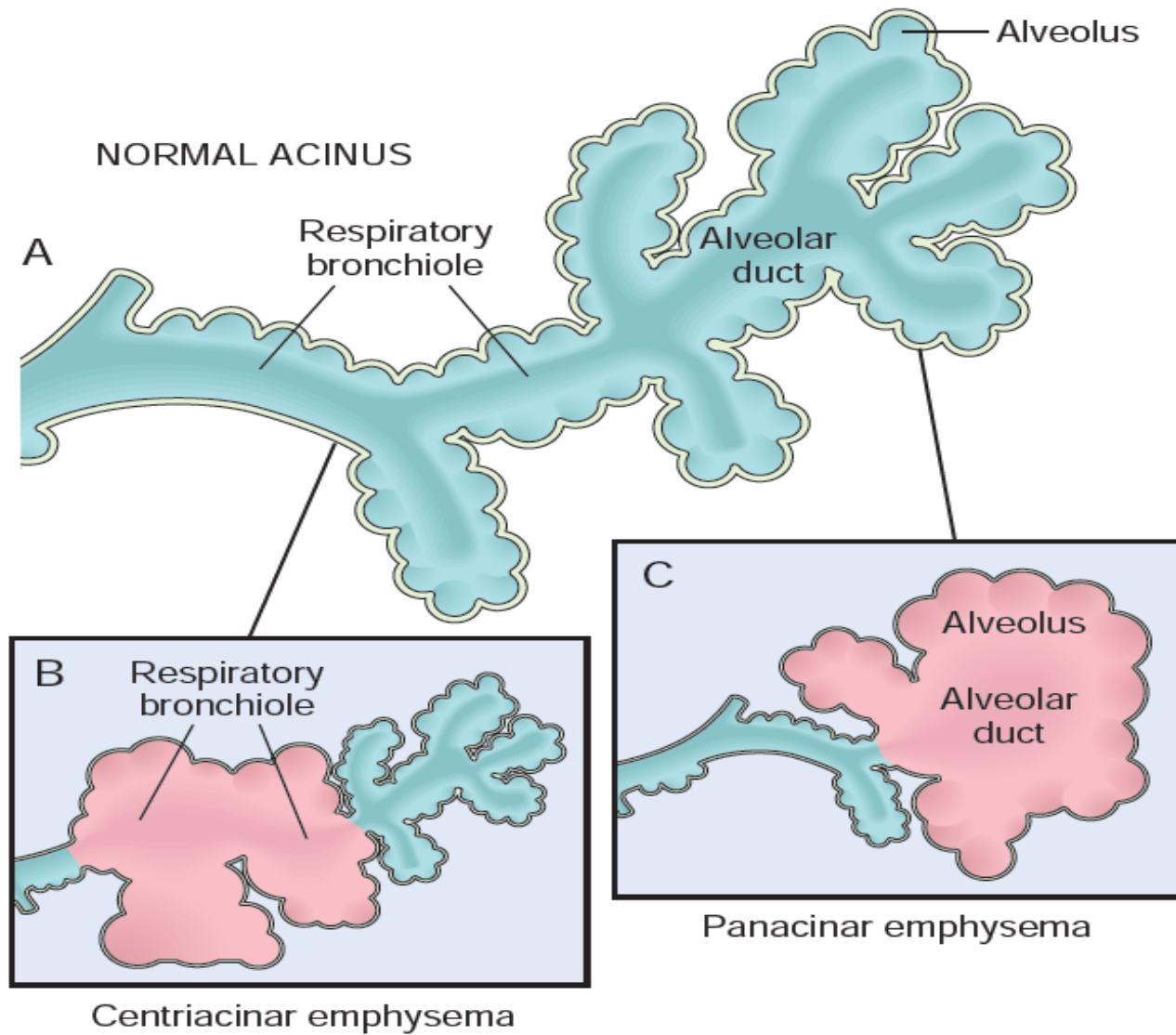
b. Panacinar (Panlobular) Emphysema

- ✓ Occurs in **α 1 -antitrypsin deficiency**
- ✓ **basal lobes**

c. Distal Acinar (Paraseptal) Emphysema

- ✓ Is more striking adjacent to the pleura
- ✓ Adjacent to fibrosis, scarring or atelectasis.
- ✓ Characterized by presence of multiple, enlarged air spaces ranging from less than 0.5 mm to more than 2.0 cm, that may form large cysts called bullae.
- ✓ It is the most common cause of **spontaneous pneumothorax** in young adults.

d. Irregular emphysema



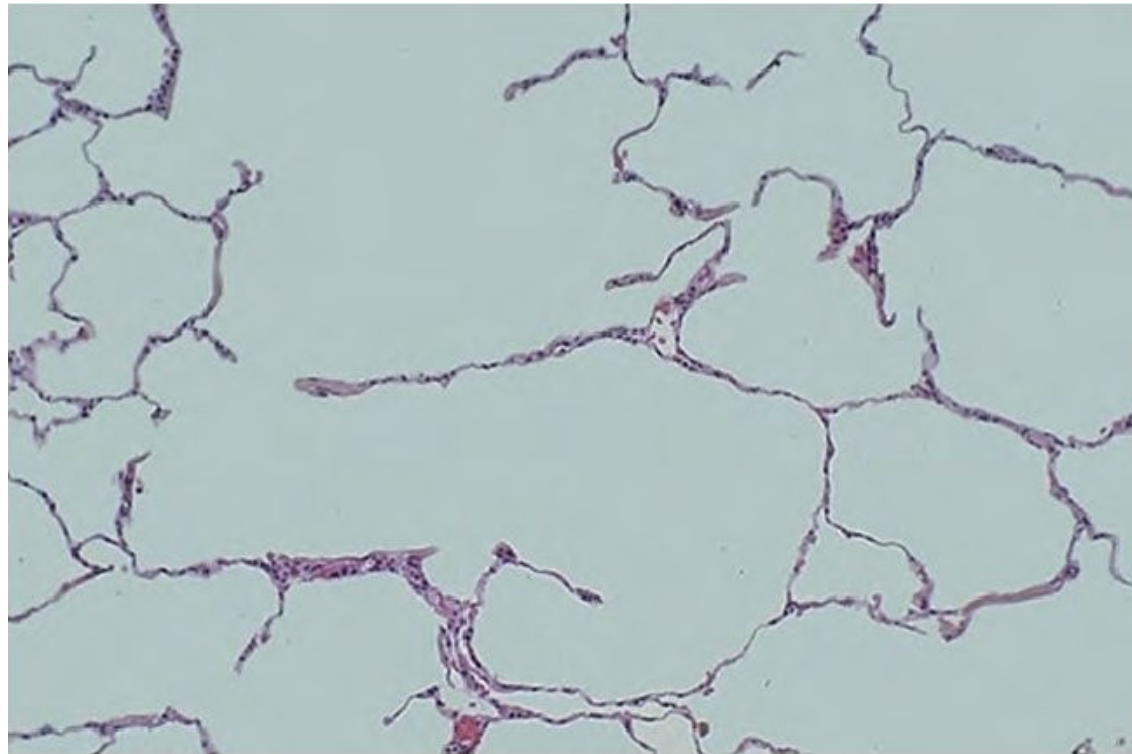
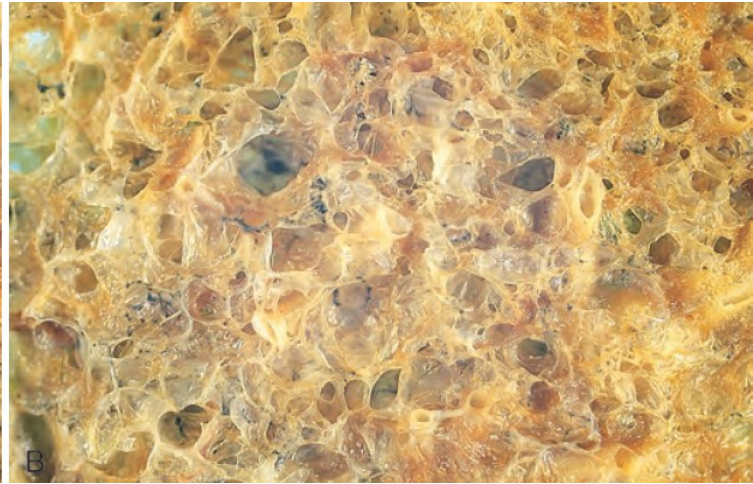
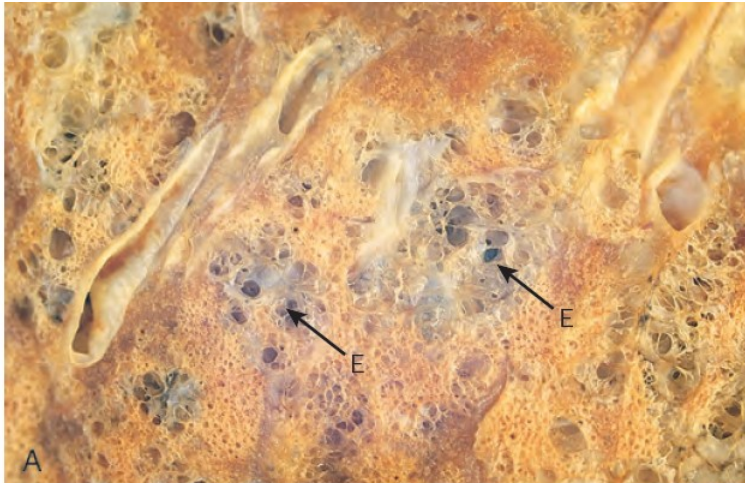


Figure 5-23 Pulmonary emphysema, Microscopic There is loss of distal airspaces: bronchioles, alveolar ducts, and alveoli. The remaining airspaces become dilated as shown here; overall, there is less surface area for gas exchange.

CLINICAL COURSE

- Dyspnea begins insidiously but is steadily progressive!
- The classical presentation is that the patient is **barrel chested** and dyspneic, sitting forward in a **hunched-over** position, attempting to squeeze the air out of the lungs with each expiratory effort, with an obviously **prolonged expiration**.
- Hyperventilation is prominent, so in early disease the gas exchange is **adequate**.
- Overventilation helps them remain well oxygenated (**pink puffers**).
- With time pulmonary hypertension develops.



2. CHRONIC BRONCHITIS

- Defined by the presence of a **persistent productive cough** for **at least 3 consecutive months** in **at least 2 consecutive years**.
- The single most important cause is cigarette smoking,
- These environmental irritants induce
 - ✓ **hypertrophy of mucous glands** in the trachea and bronchi
 - ✓ increase in mucin-secreting goblet cells in the epithelial surfaces of smaller bronchi and bronchioles
 - ✓ inflammation
- The distinctive feature of chronic bronchitis is **hypersecretion of mucus**, beginning in the large airways.
- Infiltrates of neutrophils, macrophages, and CD8+ T cells (**IL-13** release stimulates mucus secretion)

- Airflow obstruction in chronic bronchitis results from:

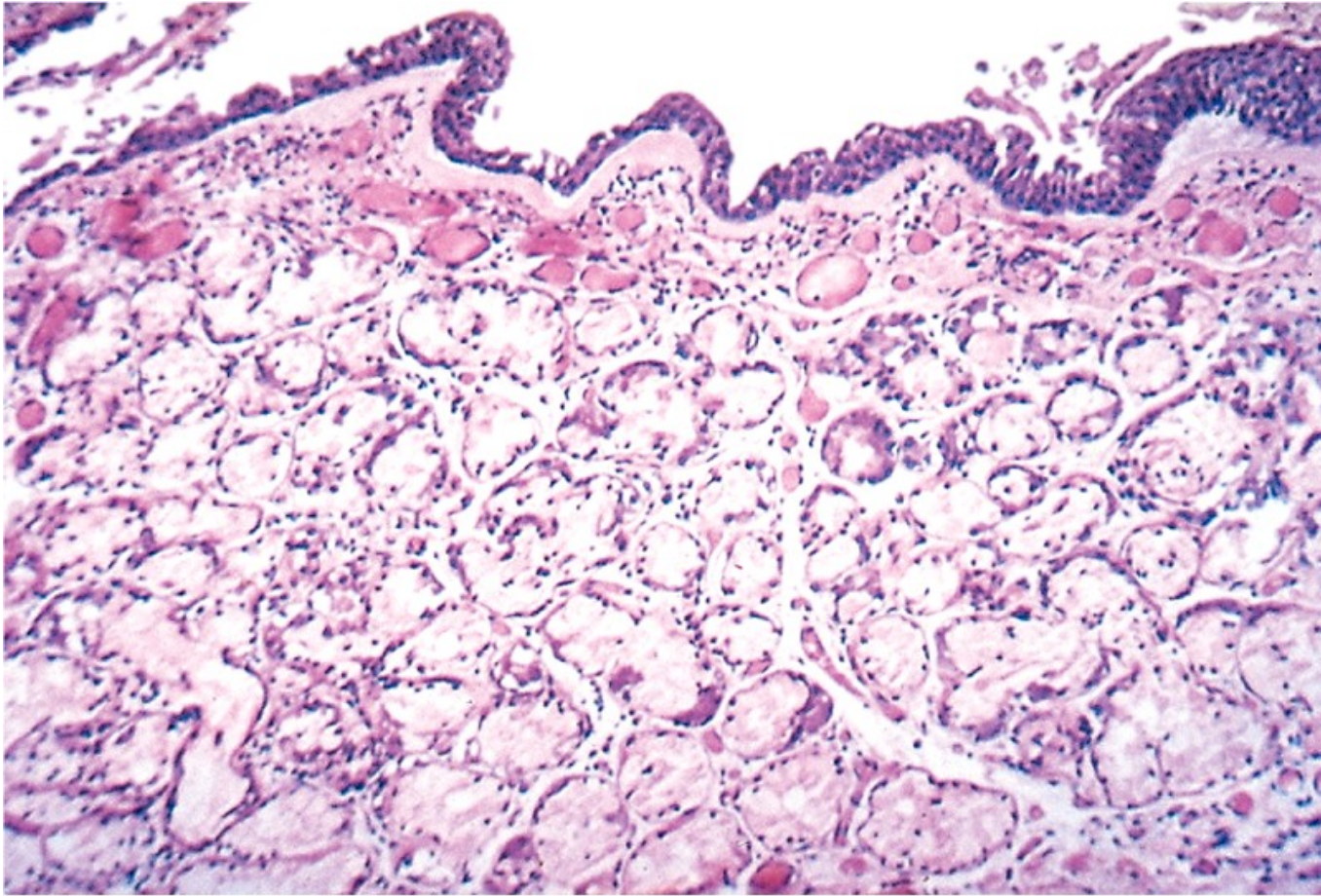
1. **Small airway disease(chronic bronchiolitis)**

- ✓ is component of early and relatively mild airflow obstruction.
- ✓ Induced by mucus plugging of the bronchiolar lumen, inflammation, and bronchiolar wall fibrosis
- ✓ Mucus plugging may completely occlude bronchiolar airways (**bronchiolitis obliterans**) and often results in opportunistic infections

2. **Coexistent emphysema**

- ✓ The cause of significant airflow obstruction

Chronic Bronchitis



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CLINICAL FEATURES:

- Prominent cough with **production of sputum**
- Some patients develop significant outflow obstruction marked by hypercapnia, hypoxemia, and cyanosis (term "blue bloaters")
- Patients may have frequent exacerbations, more rapid disease progression, and poorer outcomes than those with emphysema alone
- Progressive disease is marked by the development of pulmonary hypertension, cardiac failure, recurrent infections, and ultimately respiratory failure.

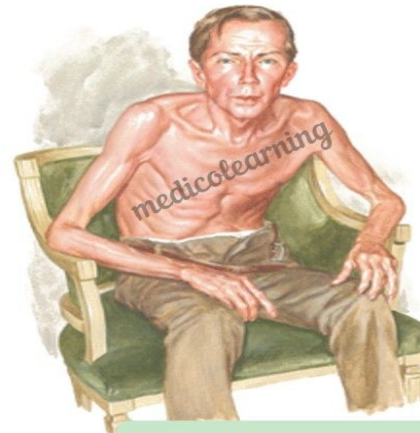
Blue Bloater (Chronic Bronchitis)



Symptoms

- Chronic, productive cough
- Purulent sputum
- Hemoptysis
- Mild dyspnea initially
- Cyanosis (due to hypoxemia)
- Peripheral edema (due to cor pulmonale)
- Crackles, wheezes
- Prolonged expiration
- Obese
- Complications
- Secondary polycythemia vera due to hypoxemia
- Pulmonary hypertension due to reactive vasoconstriction from hypoxemia
- Cor pulmonale from chronic pulmonary hypertension

Pink Puffer (Emphysema)



Symptoms

- Dyspnea
- Minimal cough
- Increased minute ventilation
- Pink skin, Pursed-lip breathing
- Accessory muscle use
- Cachexia
- Hyperinflation, barrel chest
- Decreased breath sounds
- Tachypnea
- Complications
- Pneumothorax due to bullae
- Weight loss due to work of breathing

3. ASTHMA

- Is a chronic inflammatory disorder of the airways that causes Recurrent episodes of wheezing, Dyspnea, chest tightness and cough particularly and its hallmarks are:
 - a. **Intermittent** and **reversible** airway obstruction (**bronchospasm**)
 - b. Chronic bronchial inflammation with **eosinophils**
 - c. Bronchial **smooth muscle cell hypertrophy** and **hyperreactivity**
 - d. **increased mucus secretion**

TYPES OF ASTHMA:

- I. **Atopic Asthma** : Is the most common type
 - a. Usually beginning in **childhood**
 - b. Positive **family history** of allergy
 - c. The asthmatic attacks are preceded by allergic rhinitis, urticaria, or eczema
 - d. The disease is triggered by **environmental antigens**, such as dusts, pollen
 - e. Skin test with the antigen results in an immediate wheal-and-flare reaction

II. Non-Atopic Asthma

- a. negative skin test
- b. A positive family history of asthma is less common.
- c. Exposure to viral infections and air pollutants.

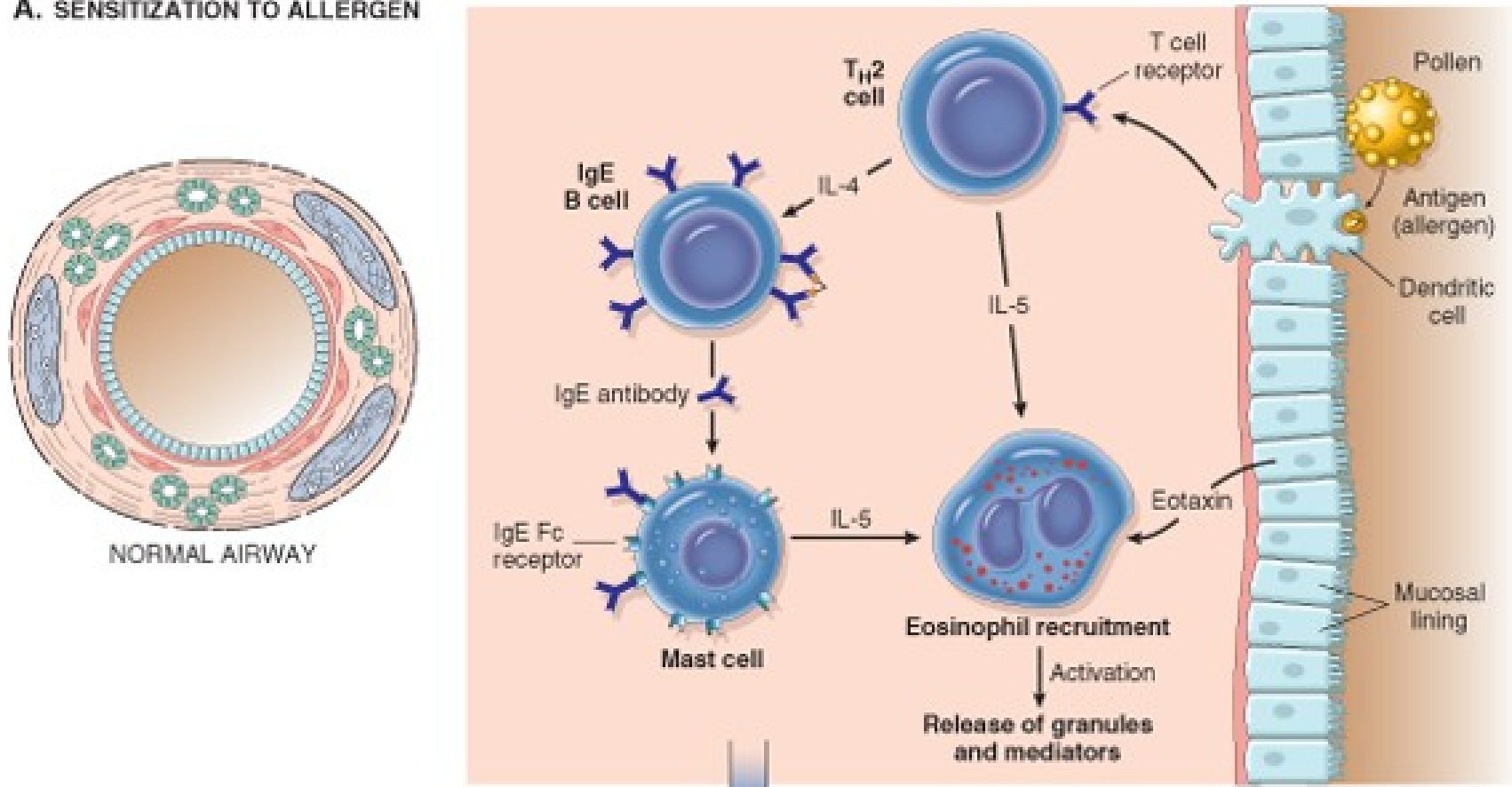
III. Drug-Induced Asthma:

- ✓ Aspirin induced asthma and patients with aspirin sensitivity
- ✓ present with recurrent rhinitis , nasal polyps , urticaria, and bronchospasm

Asthma pathogenesis

- **Sensitization phase:** antigens induce T_H2 cells
 - T_H2 cells secrete factors that stimulate IgE production by Plasma Cells
 - IgE binds to receptors on Mast Cells
 - Mast Cells produce factors that recruit Eosinophils to tissue

A. SENSITIZATION TO ALLERGEN

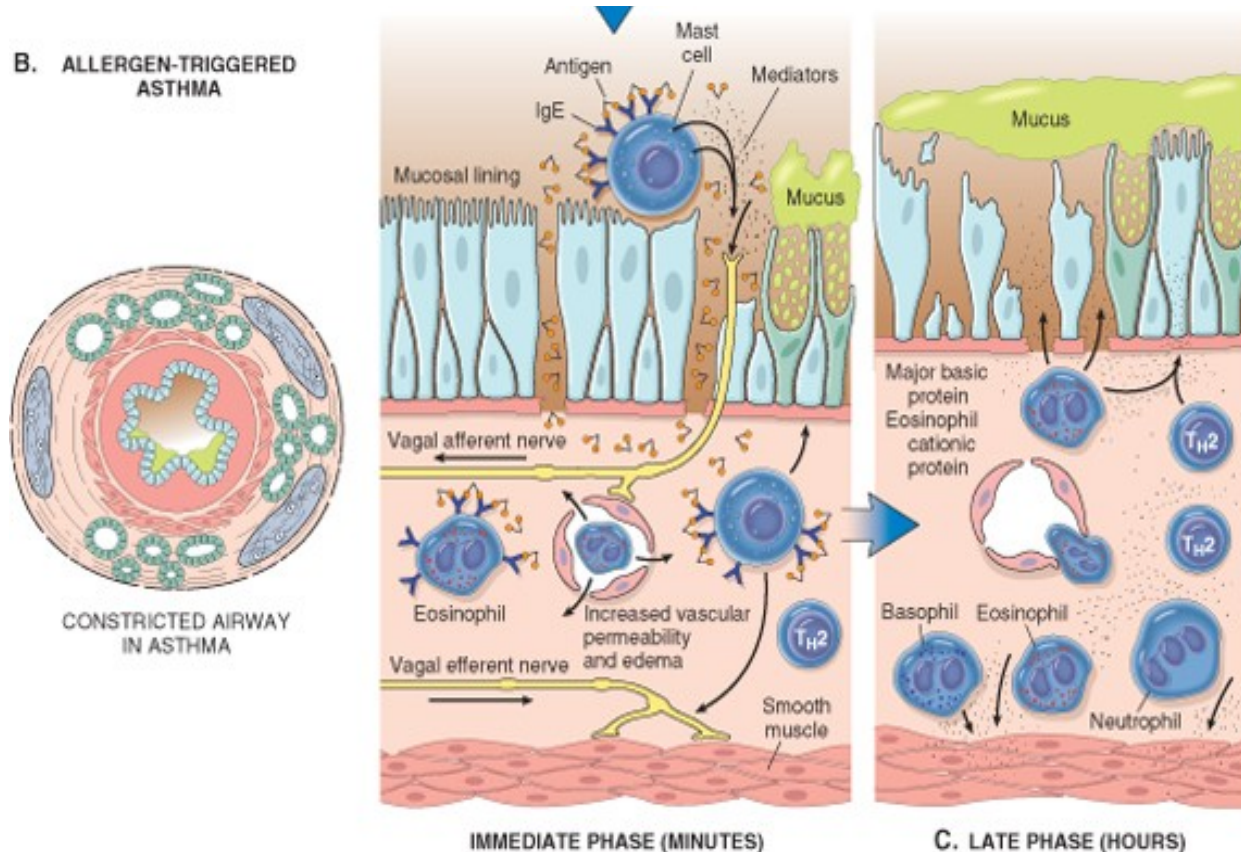


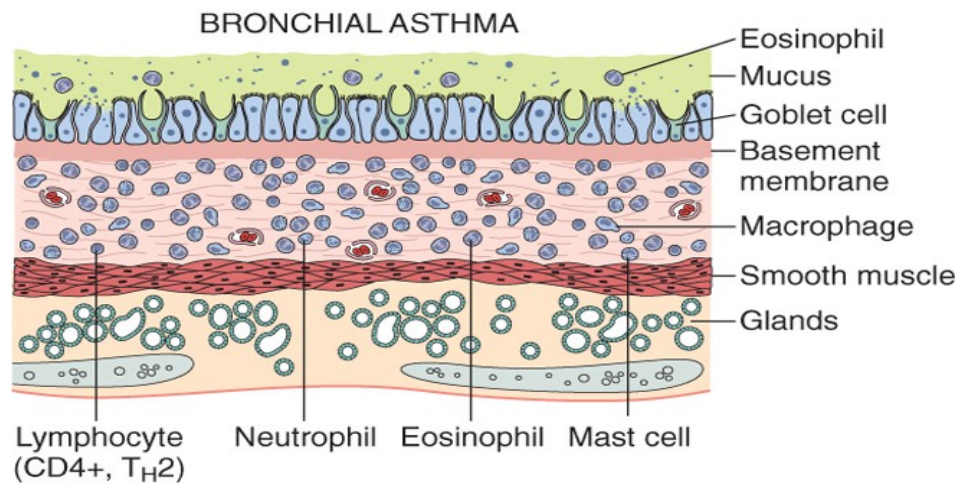
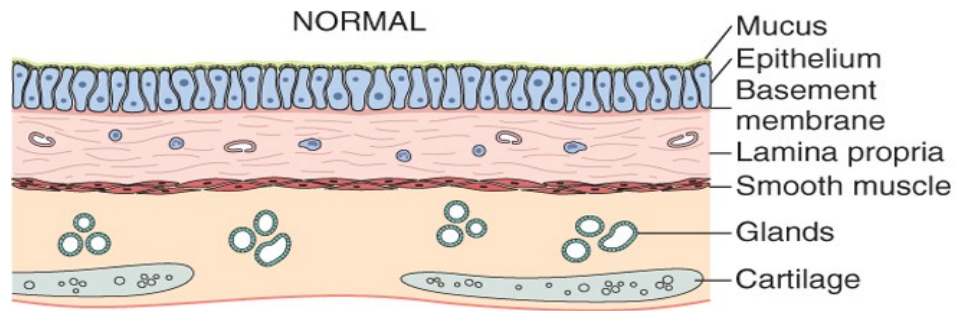
Asthma pathogenesis

Reactive phases:

Immediate: antigen binding to IgE on **Mast Cells** stimulates degranulation, releasing factors (e.g. leukotrienes, prostaglandin, histamine) that cause **bronchoconstriction**, **edema**, and **mucus secretion** AND recruit more inflammatory cells (particularly eosinophils).

Late: Eosinophils degranulate, releasing factors that damage bronchial tissue





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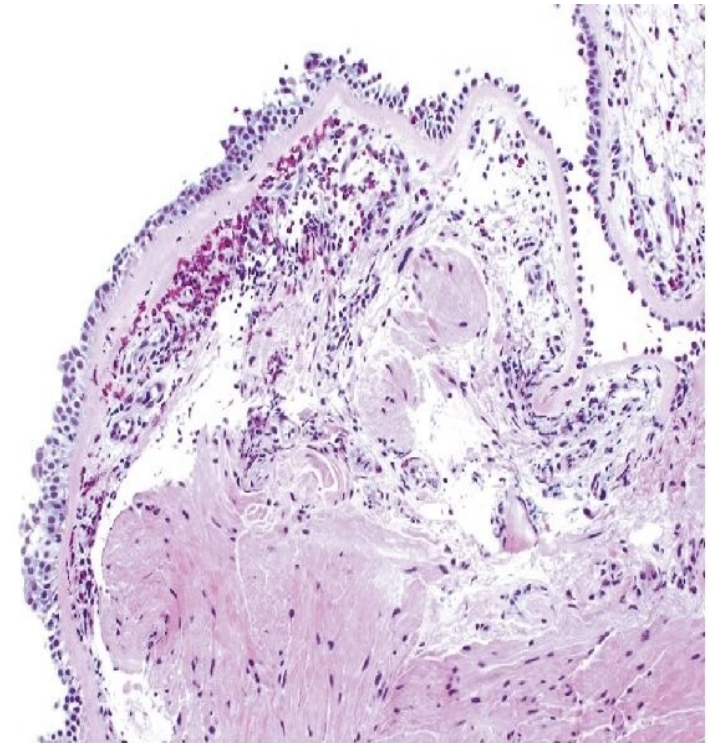


Fig. 13.11 Bronchial biopsy specimen from an asthmatic patient showing sub basement membrane fibrosis, eosinophilic inflammation, and smooth muscle hyperplasia

CLINICAL FEATURES

- Characterized by dyspnea with wheezing; and the chief difficulty lies in **expiration** so the victim labors to get air into the lungs and then cannot get it out, **progressive hyperinflation** of the lungs.
- In the usual case, attacks last from 1 to several hours and subside either **spontaneously** or with **therapy**
- Intervals between attacks are **free** from respiratory difficulties
- Occasionally a severe paroxysm occurs that **does not respond** to therapy and persists for days or weeks (**status asthmaticus**)
- The associated hypercapnia, acidosis, and severe hypoxia may be **fatal**.

BRONCHIECTASIS

- Bronchiectasis is the permanent dilation of bronchi and bronchioles caused by **destruction of the muscle and elastic supporting tissue**
- results from or is associated with chronic necrotizing infections.
- Usually affects **lower lobes**
- Characteristic symptom complex dominated by cough and expectoration of copious amounts of purulent sputum.
- Characterized by **markedly dilated bronchi** and **distal airways** and **intense acute inflammation** within bronchi and bronchioles

CAUSES:

- **Bronchial obstruction.** Common causes are
 - ✓ tumors, foreign bodies, and occasionally impaction of mucus (localized)
- Atopic asthma and chronic bronchitis
- Congenital or hereditary conditions.
 - In cystic fibrosis (widespread)
- immunodeficiency states
- Kartagener syndrome
- Necrotizing or suppurative pneumonia, particularly with virulent organisms such as **Staphylococcus aureus** or **Klebsiella**

- A persistent necrotizing inflammation in the bronchi or bronchioles may cause obstructive secretions, inflammation throughout the wall (with peribronchial fibrosis and traction on the walls).
- In cases of severe, widespread bronchiectasis, significant obstructive ventilatory defects develop, with hypoxemia, hypercapnia, pulmonary hypertension, and (rarely) cor pulmonale

Bronchiectasis

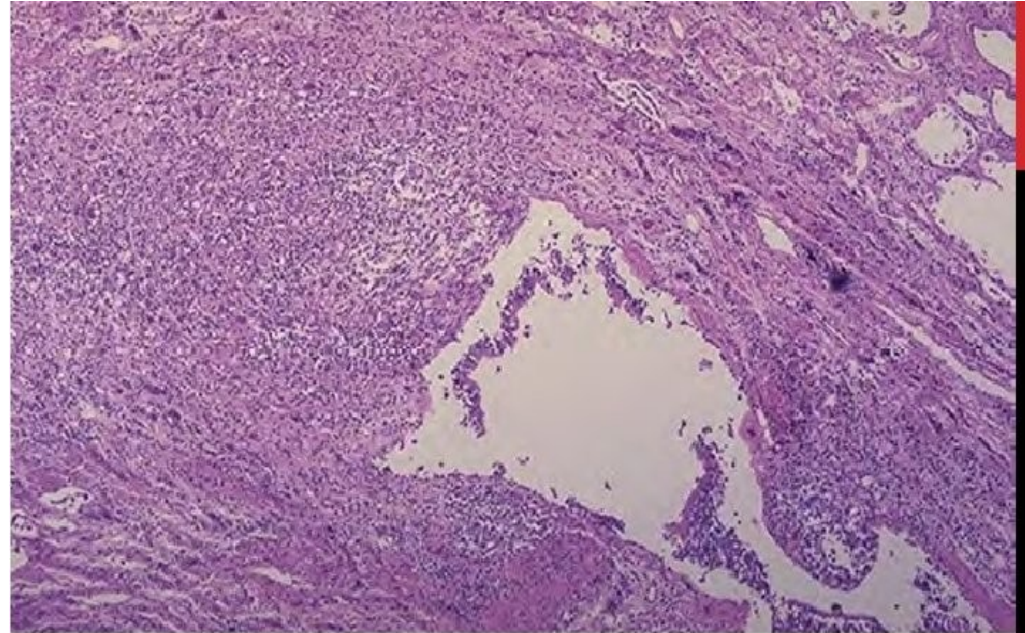
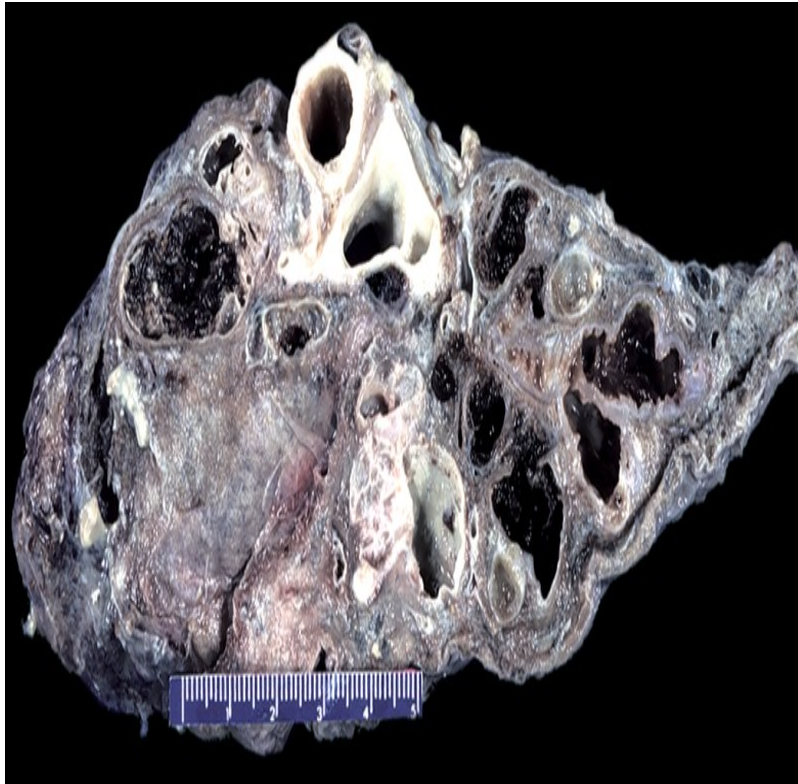


Figure 5-34 **Bronchiectasis**, **microscopic** dilated bronchus in which the mucosa and bronchial wall are not seen clearly because of the necrotizing inflammation with tissue destruction. Bronchiectasis is not a specific disease, but a consequence of another disease process that destroys airways.

ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS)

- **Clinical syndrome** caused by diffuse alveolar capillary and epithelial damage (DAD)
- Characterized by :
 - a. Rapid onset of respiratory insufficiency
 - b. dyspnea
 - c. Cyanosis
 - d. Severe arterial hypoxemia that becomes refractory to oxygen therapy and may progress to multisystem organ failure

CAUSES:

- Respiratory distress syndrome of the newborn is caused by a primary deficiency of **surfactant**.
- Causes of ARDS in adults:
 1. Direct lung injury: atypical pneumonia and aspiration of gastric content
 2. Indirect lung injury: sepsis , shock, trauma

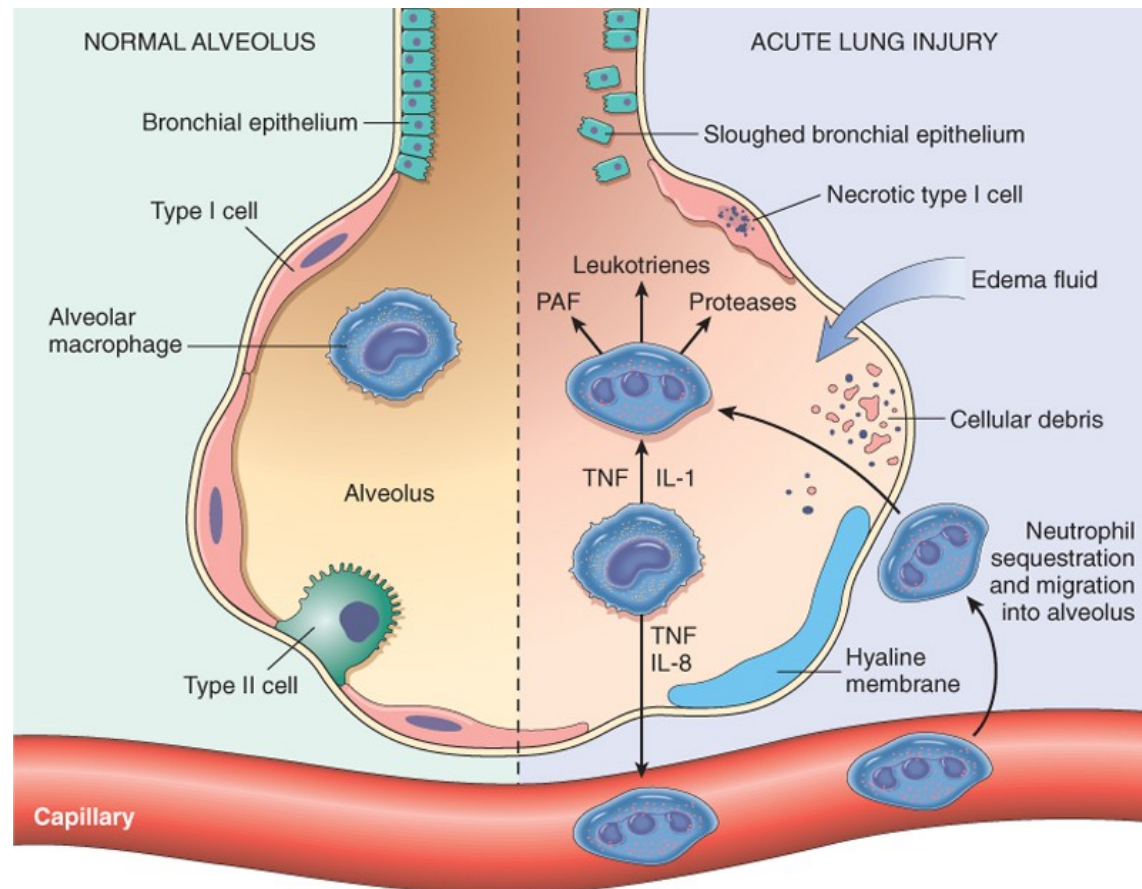
Predictors of poor prognosis include :

- a. Advanced age with Underlying sepsis ,
 - b. Development of multisystem failure
- However, in most patients who survive the acute insult and are spared the chronic sequelae, normal respiratory function returns within 6 to 12 months

ARDS:

Acute Respiratory Distress Syndrome

- Imbalance in pro- and anti-inflammatory mediators
 - NF- κ B induces macrophages to release TNF, IL-8, & IL-1 which recruits and activates neutrophils
 - Activated neutrophils release factors that destroy tissue and recruit MORE inflammatory cells (positive feedback loop)
- Results in diffuse damage to capillary and alveolar epithelium
 - Effusion of blood and plasma due to capillary leakage
 - Loss of surfactant due to type II pneumocyte damage



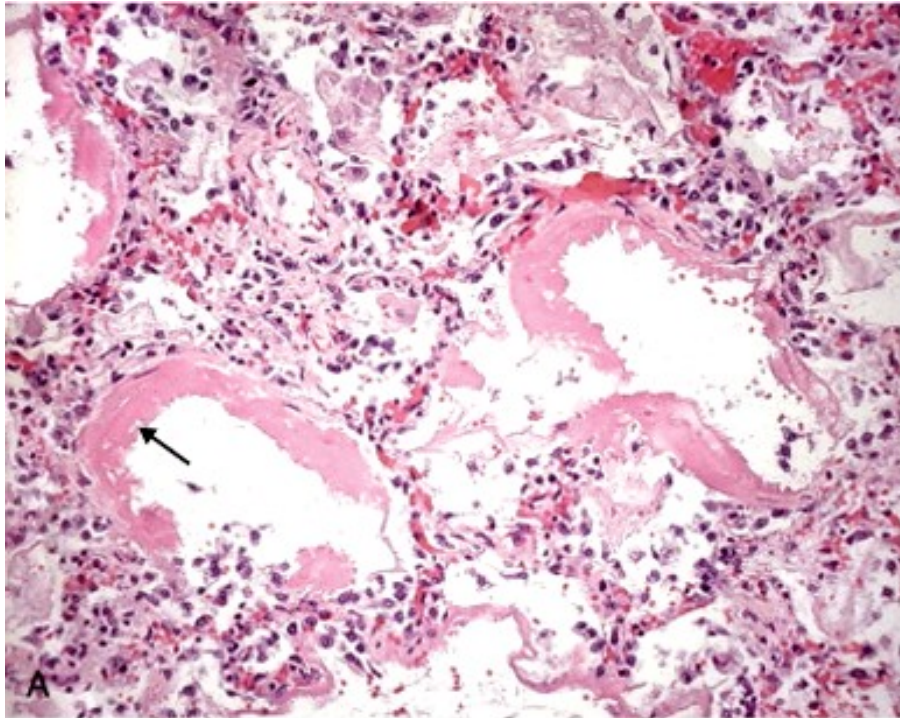
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ARDS:

Acute Respiratory Distress Syndrome

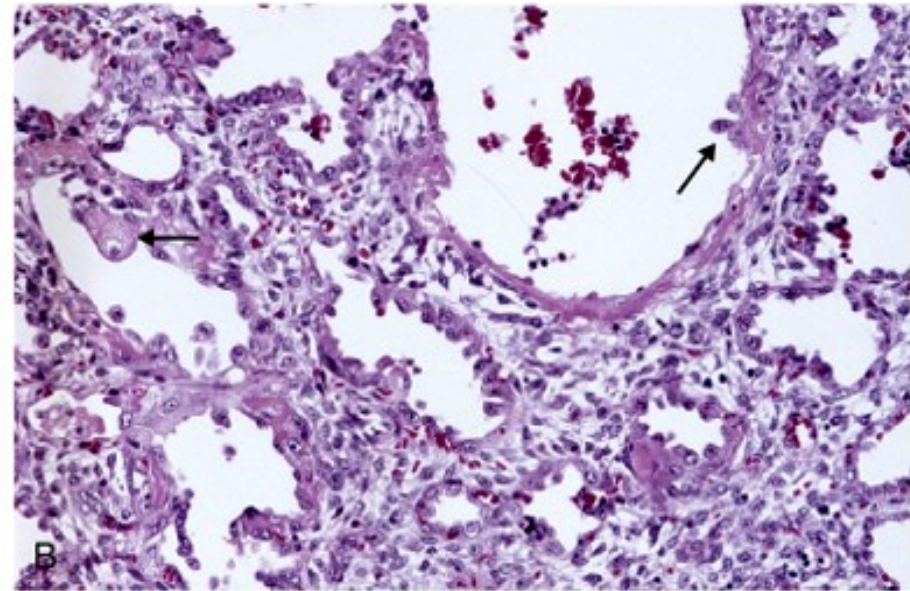
Acute Phase:

- Capillary congestion
- Interstitial and intra-alveolar edema,
- Necrotic alveolar tissue
- **Hyaline membranes:** fibrin-rich edema fluid mixed with dead cells



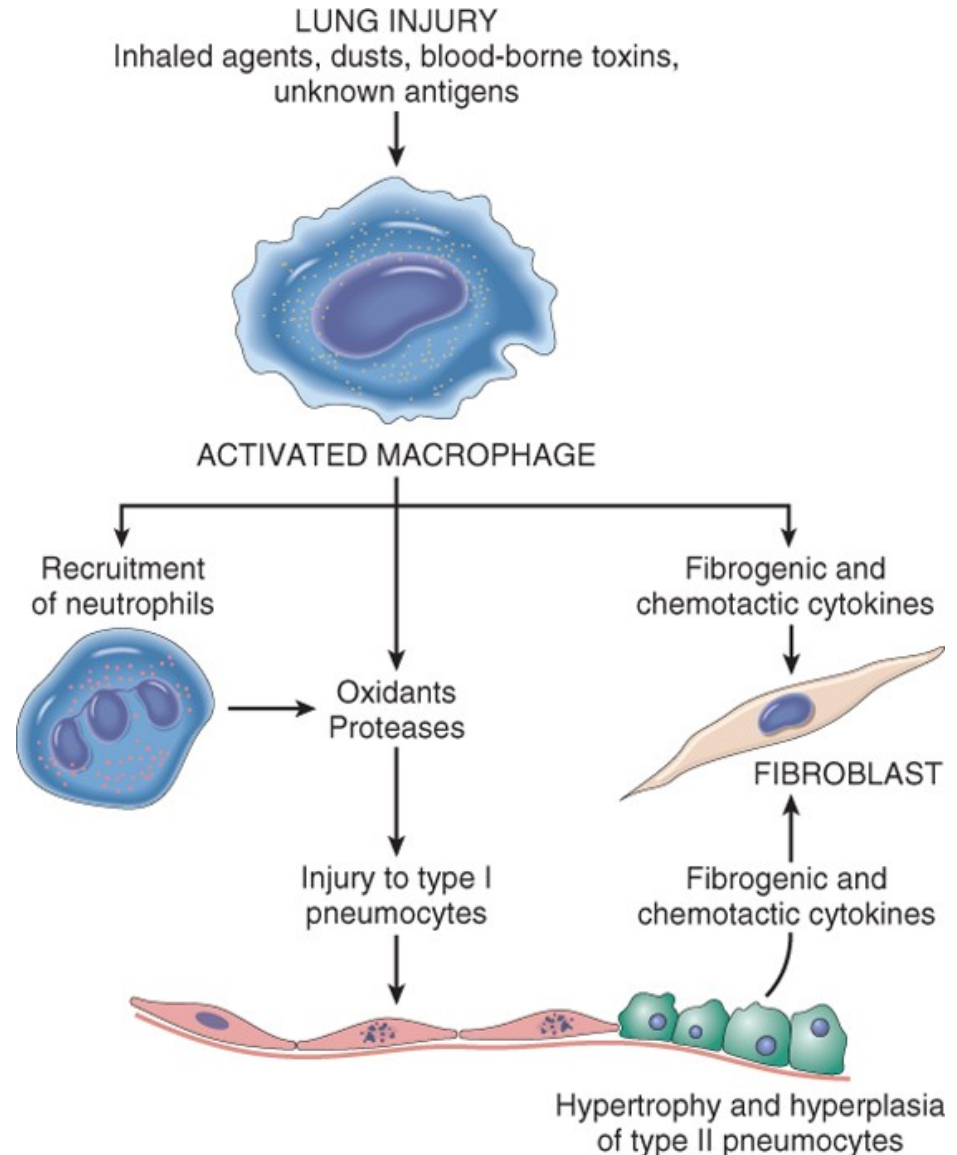
Organizing Phase:

- Proliferation of Type II pneumocytes
- Organization and fibrosis
- Thickening of alveolar septa due to proliferating interstitial cells



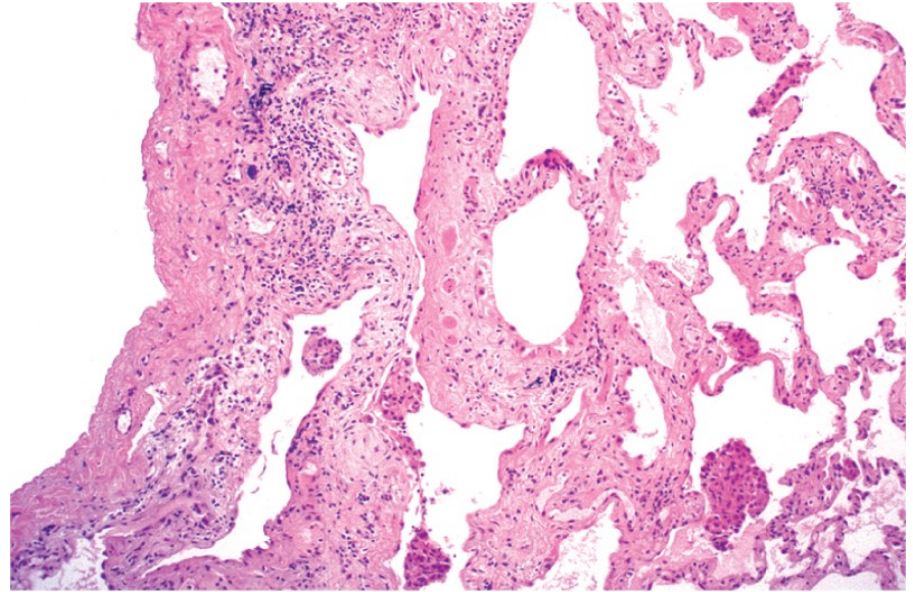
Diffuse Interstitial (Restrictive) Lung Diseases

- Diffuse, chronic fibrosis of pulmonary connective tissue (i.e. the “interstitium”) surrounding alveoli
- Stiffens and thickens lung tissue thereby reducing the airspace (FVC) AND the expiratory flow rate (FEV_1) proportionately
- Activated macrophages are the key player:
 - Damage alveoli and recruit neutrophils
 - Secrete factors that induce fibrosis
 - Damaged alveoli also induce fibrosis
- Prototypical examples are
 - ✓ **Idiopathic Pulmonary Fibrosis,**
 - ✓ **Pneumoconioses, and**
 - ✓ **Sarcoidosis**

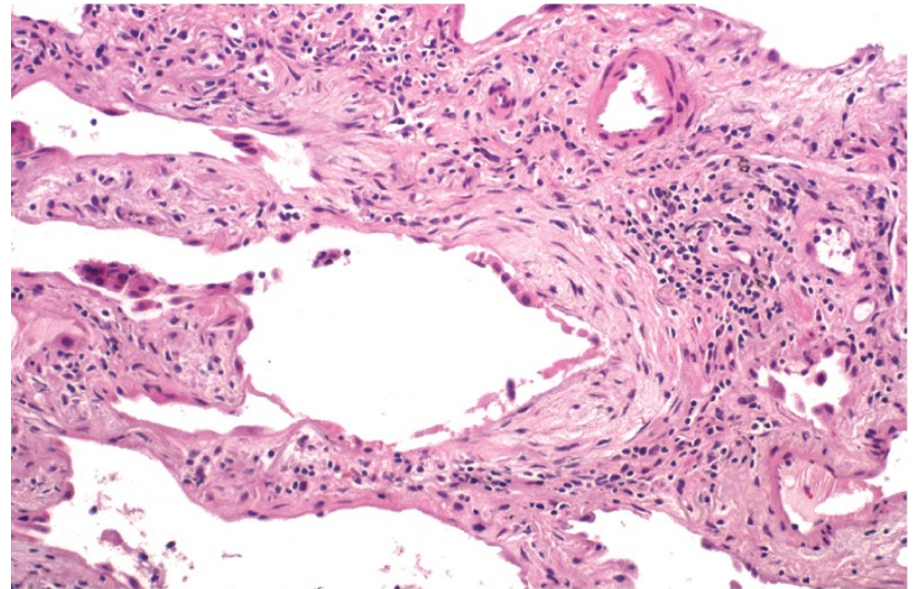


Idiopathic Pulmonary Fibrosis

- Unknown etiology (hence idiopathic)
- M > F, mostly > 60 years old
- Presents with dyspnea and non-productive cough and progresses relentlessly
- Characterized by patchy interstitial fibrosis
- Eventually collapses alveoli and large, cystic spaces form giving the lung a **honeycomb** appearance



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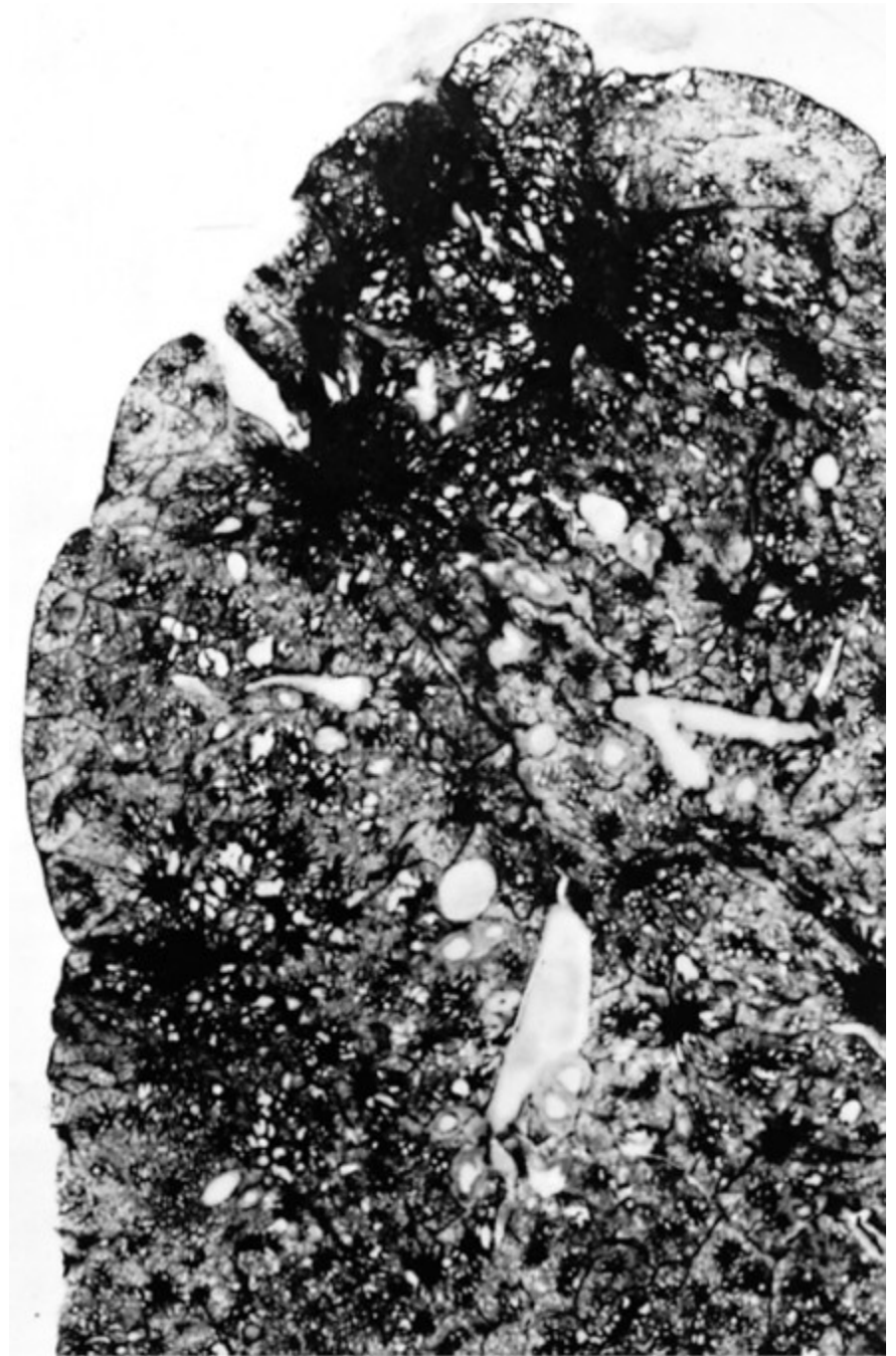
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Pneumoconioses

- Chronic fibrosing disease due to chronic exposure to particulates
- Pulmonary macrophages play central role by promoting inflammation, alveolar damage, and fibrosis
- Can progress to pulmonary dysfunction, pulmonary hypertension, and right-sided CHF
- Examples include anthracosis (black lung), silicosis, and asbestosis

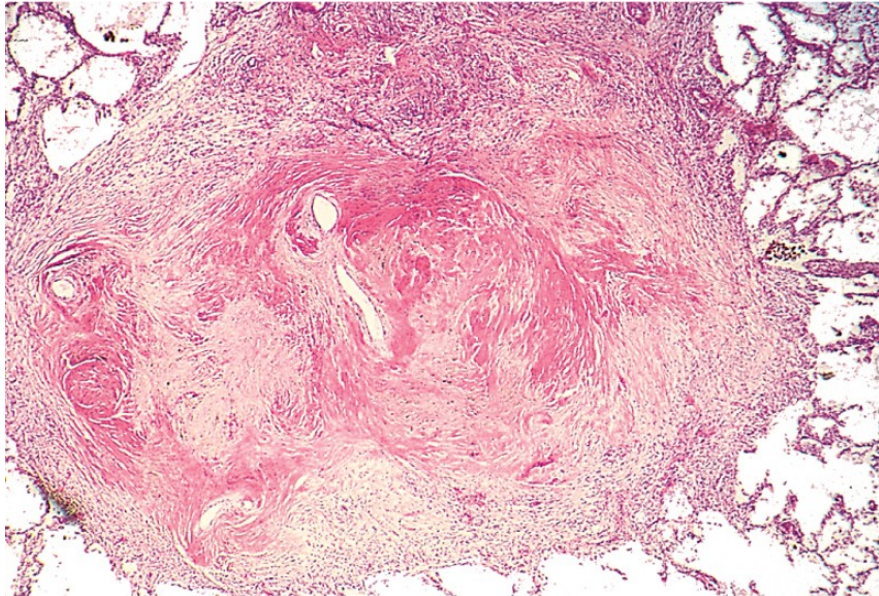
Anthracosis (black lung)

- Chronic exposure to **coal dust**
- Tends to affect upper lobes first



Silicosis

- Chronic exposure to silica (sandblasting, drywall compound)
- Characterized by silicotic nodules
- Tends to affect **upper lobes** first



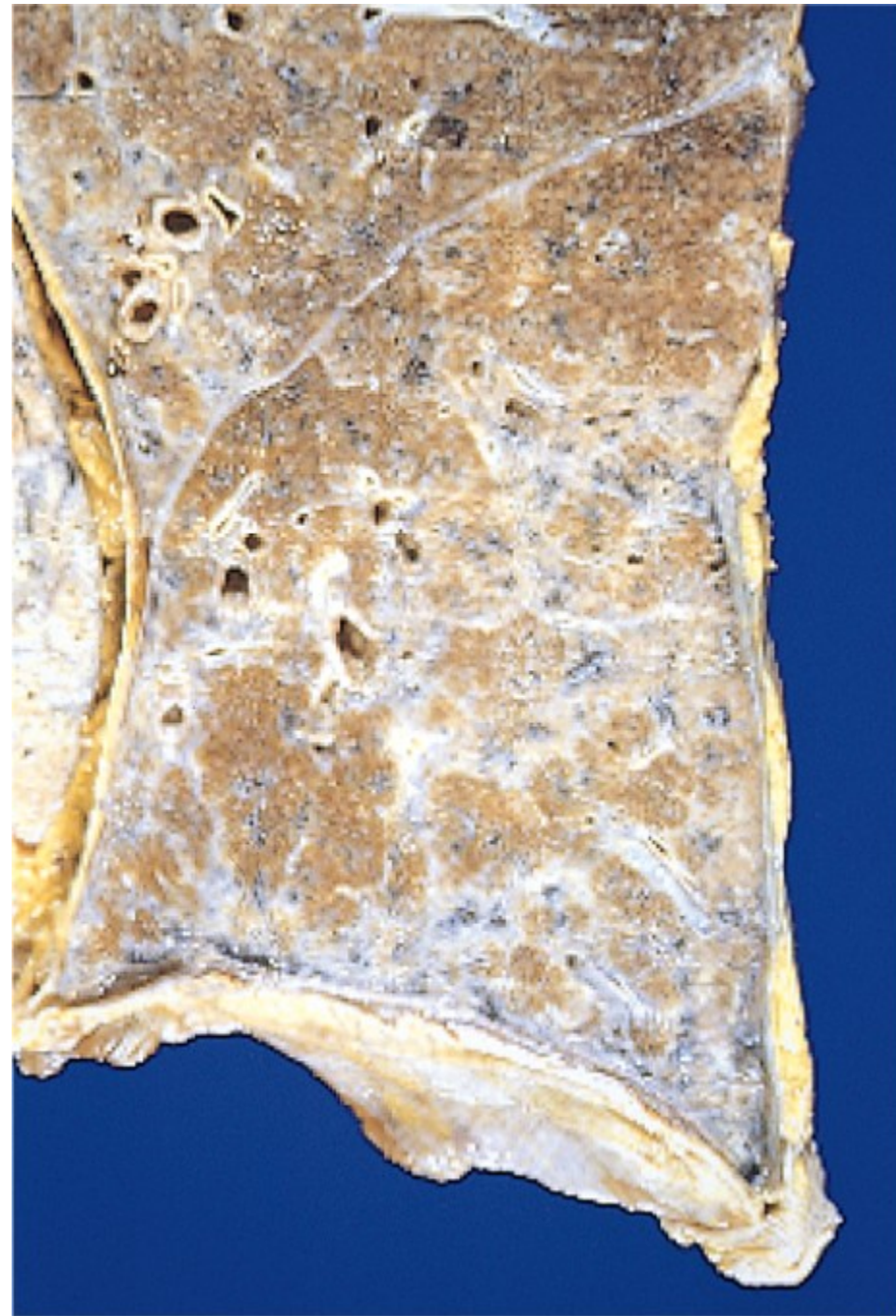
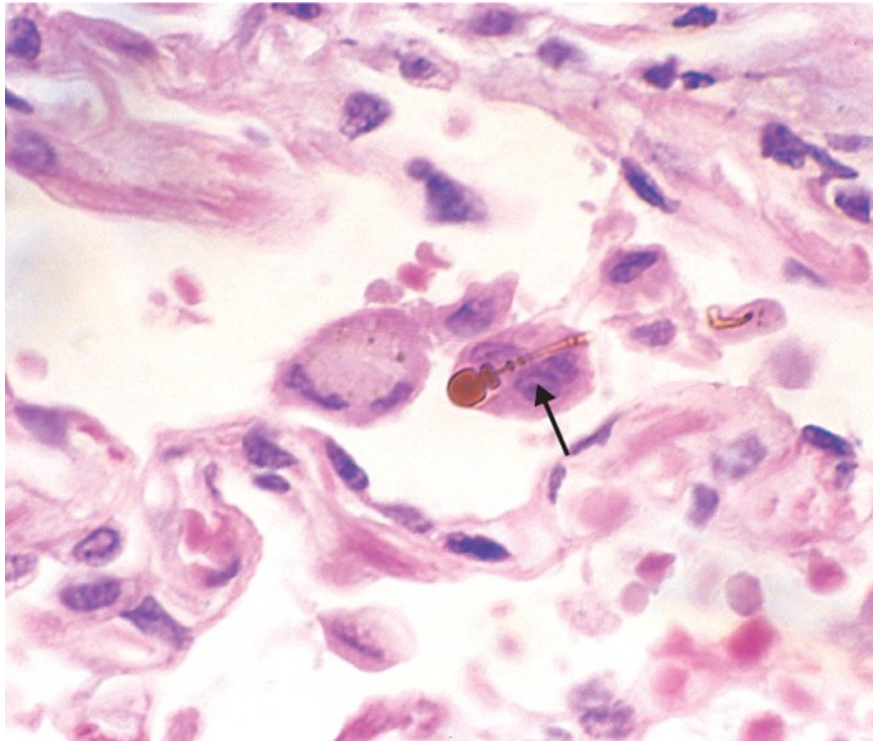
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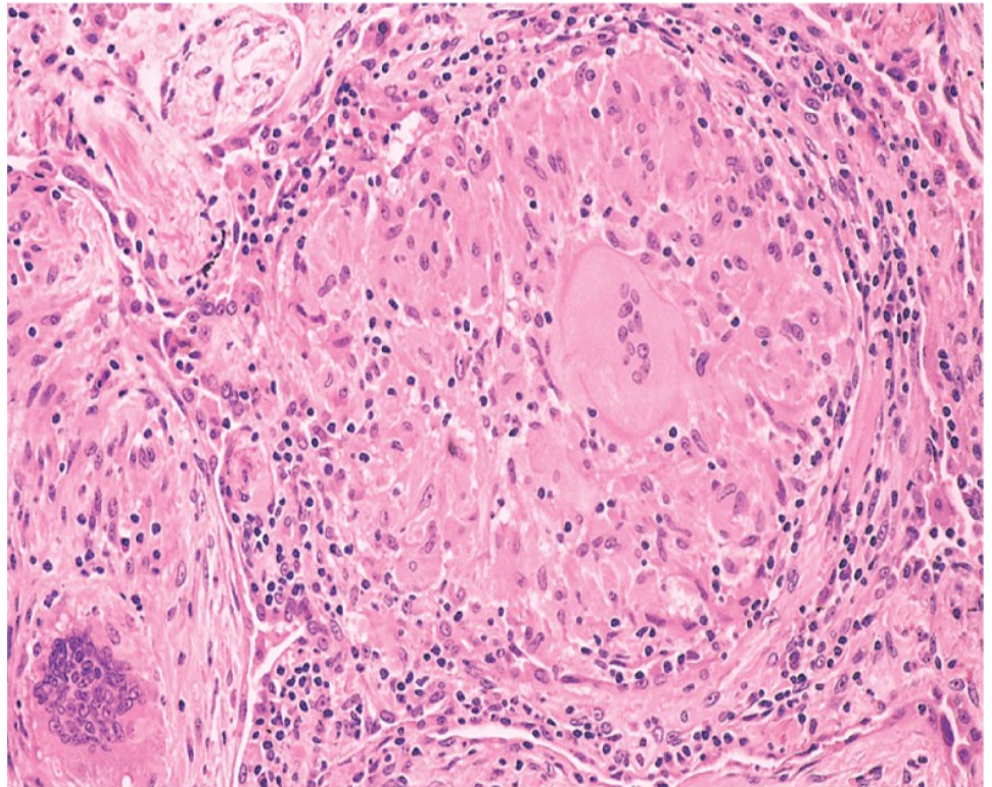
Asbestosis

- Chronic exposure to asbestos
- Characterized by asbestos bodies in the lung and fibrotic plaques in the parietal pleura
- Tends to affect middle and lower lobes



Sarcoidosis

- Multisystemic disease of unknown etiology
- Higher incidence in Scandinavian and African-American population, usually <40 yr
- Lung tissue contains high levels of CD4+ T_H1 cells and associated cytokines (IFN γ & IL-2)
- Characterized by formation of **non-caseating granulomas** and **interstitial fibrosis**



PNEUMONIA

- can be broadly defined as any infection in the lung and the clinical presentation may be as an acute fulminant clinical disease or as a chronic disease with a more protracted course.

A. Community acquired acute pneumonia

- Most are bacterial in origin and *S. pneumoniae* is the most common
- The onset usually is abrupt, with high fever, shaking chills, pleuritic chest pain and a productive mucopurulent cough;

B. Community-Acquired Atypical Pneumonias

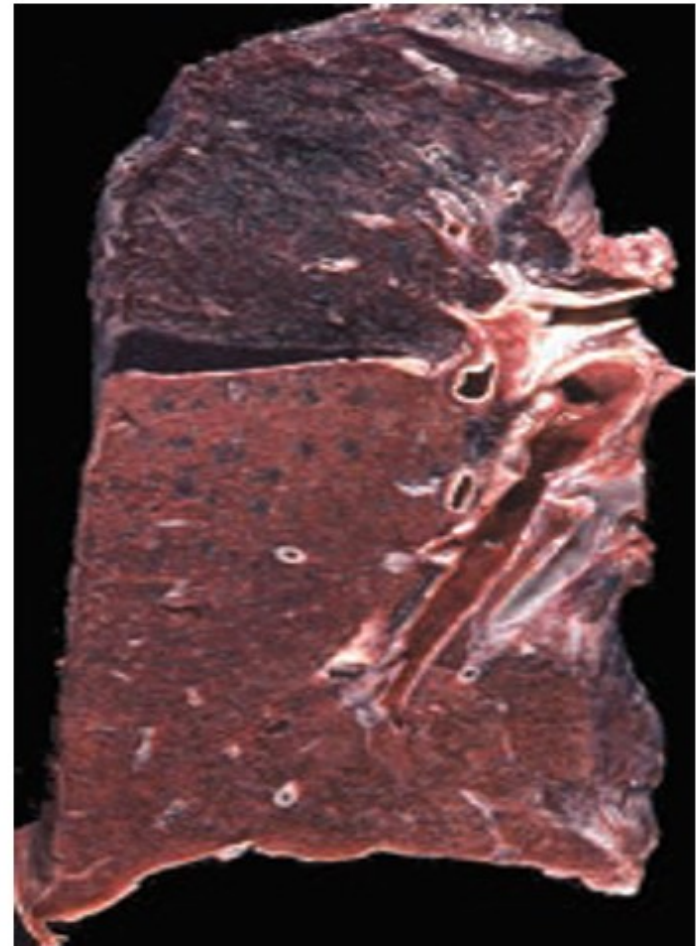
- a. The moderate amounts of sputum
 - b. Absence of physical findings of consolidation
 - c. Moderate elevation of white cell count
 - d. Lack of alveolar exudates
-
- *Mycoplasma pneumoniae* is the most common
 - particularly common among children and young adults

Morphology

- Bacterial pneumonia has two gross patterns of anatomic distribution
- ✓ **Bronchopneumonia**
- ✓ **Lobar pneumonia**
- Patchy consolidation of the lung is characteristic of bronchopneumonia
- Lobar pneumonia results in fibrinosuppurative consolidation of large portion of a lobe or of an entire lobe
- The patchy involvement may become confluent, producing total lobar consolidation
- causative agent & extent of disease – important clinically



Bronchopneumonia



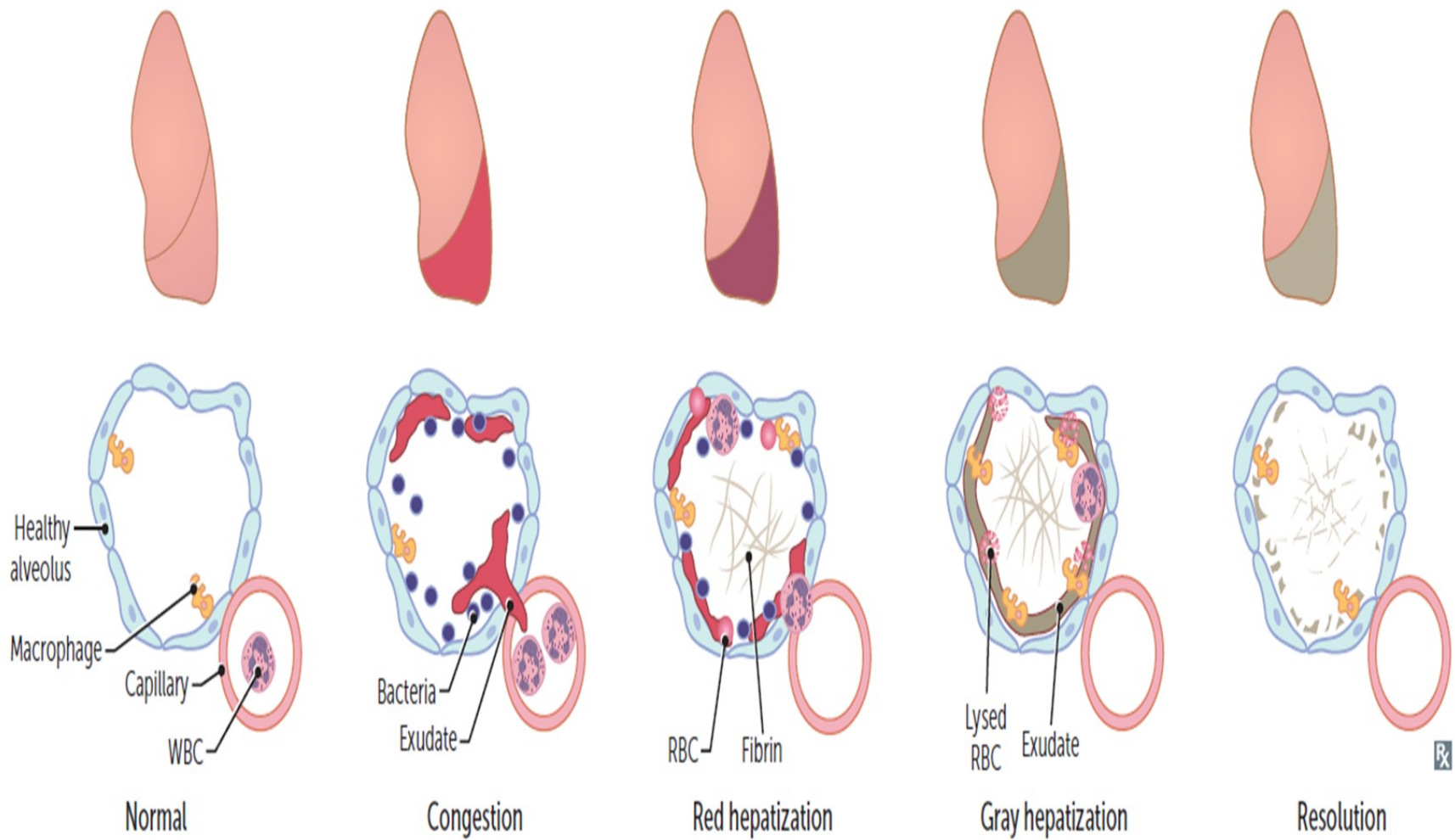
Lobar pneumonia

THE CLINICAL COURSE

- May present as a severe upper respiratory tract infection or it may manifest as a fulminant, life-threatening infection in immuno-compromised patients
- The initial presentation usually is an acute, nonspecific febrile illness characterized by fever, headache, and malaise and, later, cough with minimal sputum.

Clinical course

- In lobar pneumonia, **four stages** of the inflammatory responses have been described
 - A. Congestion:** affected lobes are congested, red, and boggy
 - B. Red hepatization:** tissue is red and firm with a liver-like consistency; alveoli are packed with red blood cells, fibrin, and neutrophils
 - C. Gray hepatization:** lung is **gray** due to lysis and ingestion of red blood cells and still **firm** and liver-like due to persistence of fibrinopurulent exudate within alveoli
 - D. Resolution:** fibrinopurulent exudate mostly resorbed by macrophages but can also organize into fibrous scar



Possible complications include:

Pleuritis: fibrinopurulent reaction in pleura may result in adhesions

Abcess formation and cavitation (loss of alveoli)

Empyema: accumulation of suppurative exudate within pleural cavity

Fibrosis

Bacteremic dissemination leading to meningitis, arthritis, or endocarditis